

Diagonalization of Defective Matrices via the Jordan Canonical Form

Theoretical Development and Practical Intuition

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Abstract

Diagonalizing a square matrix is one of the canonical procedures in computational mathematics. When a matrix possesses a complete set of linearly independent eigenvectors, a suitable change of basis transforms the matrix into a diagonal matrix. In that basis, the action of a complicated linear operator is reduced to independent scalar multiplications. This simplification has important consequences in linear algebra, differential equations, dynamical systems, numerical computation, and many other areas of scientific studies.

However, not every matrix possesses enough linearly independent eigenvectors to be diagonalized. Such matrices are called *defective*. At first sight, this appears to bring the diagonalization procedure to an abrupt end: the characteristic polynomial may provide all the eigenvalues required by the dimension of the space, while the corresponding eigenspaces fail to provide enough vectors to construct an eigenvector basis.

When a familiar mathematical procedure reaches such an obstruction, a new idea is often required. The Jordan canonical form provides one of the most elegant examples of such an idea. Instead of abandoning the simplification offered by eigenvalues and eigenvectors, the theory extends the notion of an eigenvector to that of a *generalized eigenvector*. These vectors organize themselves into chains that reveal the internal structure hidden behind repeated eigenvalues. In an appropriate basis, a defective matrix can then be transformed into a matrix that is not necessarily diagonal, but is as close to diagonal as its algebraic structure permits.

The purpose of this note is to develop the Jordan canonical form from this point of view. Rather than presenting Jordan form merely as an algorithm for arranging eigenvalues and ones into blocks, we seek to understand why generalized eigenvectors arise, why Jordan chains have their particular structure, and why the resulting similarity transformation works. Particular emphasis is placed on practical intuition so that the underlying ideas remain accessible even when the algebra becomes technically demanding.

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1 Introduction

For a diagonalizable matrix $A \in \mathbb{F}^{n \times n}$, there exists an invertible matrix P such that

$$P^{-1}AP = D,$$

where D is diagonal. If

$$P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix},$$

then the columns $v_1, \dots, v_n$ form a basis of eigenvectors of A , and

$$Av_i = \lambda_i v_i.$$

Consequently,

$$D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

The importance of this representation is difficult to overstate. A matrix that may originally mix many coordinates simultaneously is replaced, in the eigenvector basis, by independent scalar actions. For example,

$$D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k),$$

and

$$e^{tD} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}).$$

Thus powers of matrices, matrix exponentials, systems of differential equations, and many questions concerning long-term dynamical behavior become considerably easier once a suitable eigenvector basis is known.

There is, however, a fundamental difficulty.

Suppose an $n \times n$ matrix has n eigenvalues when algebraic multiplicity is counted, but the corresponding eigenspaces fail to provide n linearly independent eigenvectors. We can no longer construct the matrix P entirely from ordinary eigenvectors. Diagonalization therefore fails.

This raises the central question of this note:

What should we do when the eigenvectors run out?

The Jordan canonical form provides an answer by asking us not to discard the available eigenvectors, but instead to investigate the structure surrounding them.

Suppose now that an eigenvalue λ has algebraic multiplicity $m > 1$. Ideally, we would like to find m linearly independent eigenvectors associated with λ . If this happens, the repeated eigenvalue itself causes no difficulty for diagonalization.

The difficulty arises when the eigenspace

$$E_\lambda = \ker(A - \lambda I)$$

has dimension strictly smaller than the algebraic multiplicity of λ . In other words,

$$\dim \ker(A - \lambda I) < m.$$

Then there are not enough ordinary eigenvectors associated with λ to account for all m occurrences of that eigenvalue in the characteristic polynomial.

For example, suppose that λ has algebraic multiplicity 3, but

$$\dim \ker(A - \lambda I) = 1.$$

Then we can find only one linearly independent ordinary eigenvector, say v_1 , satisfying

$$(A - \lambda I)v_1 = 0.$$

At this point ordinary diagonalization has run out of eigenvectors. However, rather than abandoning the eigenvalue λ , we ask whether there are other vectors that are connected to its eigenspace through the operator $A - \lambda I$.

In particular, we search for a vector v_2 satisfying

$$(A - \lambda I)v_2 = v_1.$$

The vector v_2 is not an ordinary eigenvector, since $(A - \lambda I)v_2 \neq 0$. Nevertheless, applying $A - \lambda I$ once sends v_2 directly into the eigenspace.

If another vector is still required, we may search for v_3 satisfying

$$(A - \lambda I)v_3 = v_2.$$

The resulting vectors form the chain

$$v_3 \xrightarrow{A-\lambda I} v_2 \xrightarrow{A-\lambda I} v_1 \xrightarrow{A-\lambda I} 0.$$

Thus,

$$(A - \lambda I)v_1 = 0,$$

$$(A - \lambda I)v_2 = v_1,$$

and

$$(A - \lambda I)v_3 = v_2.$$

The vector v_1 is an ordinary eigenvector, while v_2 and v_3 are generalized eigenvectors. Together they form what is called a *Jordan chain* associated with the eigenvalue λ .

This construction provides the missing basis vectors that ordinary eigenvectors could not supply. As we shall see, the structure of this chain is precisely what produces a Jordan block. The purpose of this note is not merely to state this construction. We will develop it gradually from the failure of ordinary diagonalization. We will examine the role of the operator

$$N = A - \lambda I,$$

study the nested null spaces

$$\ker(N) \subseteq \ker(N^2) \subseteq \ker(N^3) \subseteq \cdots,$$

and show how these spaces reveal the lengths and number of Jordan chains. This will naturally lead to generalized eigenvectors, Jordan blocks, the minimal polynomial, and finally the Jordan canonical form.

Throughout the development, theoretical arguments will be accompanied by practical interpretations and explicit examples. The objective is to make the Jordan construction feel less like a collection of algebraic rules and more like a natural response to the failure of ordinary diagonalization. Even when the mathematics becomes technically involved, the reader should be able to retain a clear picture of what the matrix is doing and why each new construction is needed.

2 Prerequisites

This note assumes that the reader has already mastered the basic theory of eigenvalues, eigenvectors, and diagonalization. In particular, the reader should be comfortable with the eigenvalue equation

$$Av = \lambda v,$$

and its equivalent homogeneous system

$$(A - \lambda I)v = 0.$$

The reader should also know how to obtain eigenvalues from the characteristic equation

$$\det(A - \lambda I) = 0,$$

and how to determine the eigenspace associated with an eigenvalue λ ,

$$E_\lambda = \ker(A - \lambda I).$$

We further assume familiarity with algebraic and geometric multiplicity. If an eigenvalue λ occurs m times as a root of the characteristic polynomial, then its algebraic multiplicity is m . Its geometric multiplicity is

$$\dim \ker(A - \lambda I).$$

For every eigenvalue,

$$1 \leq \text{geometric multiplicity} \leq \text{algebraic multiplicity}.$$

The reader should also be familiar with ordinary diagonalization. If an $n \times n$ matrix A possesses n linearly independent eigenvectors $v_1, \dots, v_n$, then

$$P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}$$

is invertible and

$$P^{-1}AP = D,$$

where D is diagonal.

Equivalently,

$$AP = PD.$$

This latter form is particularly useful because it exposes what diagonalization actually means:

$$A \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

Thus diagonalization is fundamentally a search for a basis in which the action of the linear operator becomes as simple as possible.

Finally, basic familiarity with null spaces, rank, basis, linear independence, similarity transformations, and matrix multiplication is assumed.

No previous knowledge of generalized eigenvectors, Jordan chains, minimal polynomials, or the Jordan canonical form is required. These ideas will be developed from the beginning.

3 Historical Development of the Jordan Canonical Form

The Jordan canonical form did not emerge as an isolated construction. It grew out of a much broader nineteenth-century effort to understand linear transformations through their characteristic quantities and to find coordinate systems in which those transformations take their simplest possible forms.

3.1 The Road Toward Eigenvalue Decomposition

The ideas that we now organize under the language of eigenvalues, eigenvectors, and matrix diagonalization developed gradually over a long period of time. Their origins can be traced to problems involving systems of linear equations, differential equations, quadratic forms, mechanics, and astronomy.

During the eighteenth century, mathematicians such as **Leonhard Euler** and **Joseph-Louis Lagrange** encountered what we would now recognize as eigenvalue problems while studying systems of differential equations and small oscillations in mechanical systems. In such problems, certain special modes of motion allowed a complicated coupled system to be separated into simpler independent motions.

The subject advanced considerably during the nineteenth century. **Augustin-Louis Cauchy** developed important results involving determinants, characteristic equations, and quadratic forms. These ideas helped establish the connection between characteristic roots and the simplification of linear transformations.

At the same time, the theory of determinants and linear substitutions was becoming increasingly systematic. Mathematicians were gradually recognizing a central principle: although the numerical entries used to represent a linear transformation depend on the chosen coordinates, certain underlying properties of the transformation remain unchanged.

This naturally led to the search for special coordinate systems in which a linear transformation could be represented as simply as possible.

In modern language, one of the most desirable outcomes of this search is *eigenvalue decomposition*, or diagonalization. When enough independent eigenvectors exist, the transformation can be represented by a diagonal matrix. A complicated coupled transformation is thereby reduced to a collection of independent scalar actions.

This was an extraordinarily powerful idea. But it came with an important limitation: not every

linear transformation possesses enough independent eigenvectors to permit such a complete diagonal reduction.

The question therefore became broader than diagonalization itself:

If complete diagonal reduction is impossible, what is the simplest form to which a linear transformation can still be reduced?

It was in the development of this broader problem that Camille Jordan entered the story.

3.2 Camille Jordan and Canonical Reduction

Marie Ennemond Camille Jordan (1838–1922) was a French mathematician whose work had a major influence on nineteenth-century algebra. His contributions extended well beyond the subject discussed in this note and included important work in group theory, algebra, analysis, and geometry.

Jordan was particularly interested in the theory of substitutions. In the terminology of the period, a linear substitution described what we would now naturally regard as a linear transformation represented through a system of linear expressions.

The fundamental problem was one of classification and reduction. Rather than accepting the potentially complicated original representation of a transformation, mathematicians wanted to determine whether it could be converted into a simpler standard form without losing its essential algebraic information.

Jordan developed a systematic theory for carrying this reduction much further, including precisely those situations in which a complete diagonal reduction was unavailable.

His treatment appeared prominently in his monumental 1870 work

Traité des substitutions et des équations algébriques.

The canonical structure associated with this work eventually became known as the *Jordan canonical form*, or *Jordan normal form*.

Jordan's contribution was important because it showed that the failure of diagonalization was not the end of the simplification process. Even when a transformation could not be completely decomposed into independent eigendirections, its remaining algebraic structure could still be organized into a remarkably simple and systematic canonical form.

3.3 Jordan's Place in a Larger Mathematical Development

As is often the case with major mathematical theories, the Jordan canonical form should not be understood as the product of a single mathematician working in isolation. It emerged from a long development involving characteristic equations, determinants, linear transformations, invariant quantities, and canonical reduction.

Important related work was carried out by mathematicians including Cauchy and, independently in the theory of elementary divisors, **Karl Weierstrass**. The nineteenth century was a period in which many of the ideas that would eventually become modern linear algebra were being developed simultaneously from several directions.

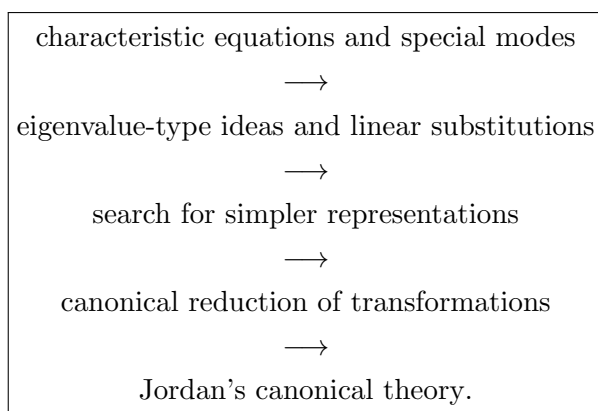
Jordan's achievement was to provide a particularly systematic canonical theory for linear substitutions. The language and notation used today came later, as matrix theory and abstract linear algebra were organized into their modern forms.

Consequently, it would be historically misleading to imagine Jordan sitting down with the modern formula

$$A = PDP^{-1}$$

and attempting to repair it when it failed. That is a useful modern way for us to motivate the subject, but it is not a literal description of its historical development.

Historically, the movement was broader:



Today, this history appears in a particularly elegant form in a first course in linear algebra. We first learn that eigenvectors can provide a basis in which a matrix becomes diagonal. We then discover that some matrices do not possess enough eigenvectors. Jordan canonical form shows that the nineteenth-century search for simplification does not stop there.

With this historical perspective in place, we now return to the mathematics. The first theoretical tool we shall need is the Cayley–Hamilton theorem.

4 Background: The Cayley–Hamilton Theorem

Before developing the theory of generalized eigenvectors and Jordan chains, we recall one result that will play an important role throughout the discussion: the *Cayley–Hamilton theorem*.

The theorem establishes a remarkable connection between a matrix and its own characteristic polynomial.

4.1 The Characteristic Polynomial

Let $A \in \mathbb{F}^{n \times n}$. We define the characteristic polynomial of A by

$$p_A(\lambda) = \det(\lambda I - A).$$

Since A is an $n \times n$ matrix, $p_A(\lambda)$ is a polynomial of degree n . It can therefore be written in the form

$$p_A(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0.$$

The eigenvalues of A are precisely the roots of this polynomial.

For example, suppose that a 3×3 matrix has characteristic polynomial

$$p_A(\lambda) = (\lambda - 2)^3.$$

Then 2 is the only eigenvalue of A , and it has algebraic multiplicity 3.

At this stage, however, the characteristic polynomial appears to be a statement about the scalar variable λ . The Cayley–Hamilton theorem tells us something much stronger: the matrix A itself satisfies this polynomial.

4.2 The Cayley–Hamilton Theorem

Theorem 4.1 (Cayley–Hamilton Theorem). *Every square matrix satisfies its own characteristic polynomial.*

More precisely, if

$$p_A(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0,$$

then

$$p_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1A + c_0I = 0.$$

Here 0 denotes the zero matrix.

The notation $p_A(A)$ simply means that the scalar variable λ in the characteristic polynomial is replaced by the matrix A . The constant term must consequently be interpreted as a multiple of the identity matrix.

For example, if

$$p_A(\lambda) = \lambda^3 - 6\lambda^2 + 12\lambda - 8,$$

then the Cayley–Hamilton theorem gives

$$A^3 - 6A^2 + 12A - 8I = 0.$$

Thus the matrix obeys the same algebraic relation as the roots of its characteristic polynomial.

4.3 A Particularly Important Repeated-Eigenvalue Case

The connection with defective matrices becomes especially interesting when the characteristic polynomial contains a repeated eigenvalue.

Suppose, for example, that

$$p_A(\lambda) = (\lambda - 2)^3.$$

The Cayley–Hamilton theorem immediately gives

$$p_A(A) = 0,$$

and hence

$$\boxed{(A - 2I)^3 = 0.}$$

More generally, if an $n \times n$ matrix has only one eigenvalue λ_0 , with algebraic multiplicity n , then

$$p_A(\lambda) = (\lambda - \lambda_0)^n,$$

and therefore

$$\boxed{(A - \lambda_0 I)^n = 0.}$$

This observation suggests introducing the shifted matrix

$$N = A - \lambda_0 I.$$

Then

$$A = \lambda_0 I + N$$

and Cayley–Hamilton tells us that

$$N^n = 0.$$

A matrix whose sufficiently high power is zero is called *nilpotent*. Thus, in this special situation, subtracting the eigenvalue from the matrix exposes a nilpotent part:

$$\boxed{A = \lambda_0 I + N, \quad N^n = 0.}$$

This decomposition will become extremely important when we develop the Jordan canonical form.

4.4 What Does $N^n = 0$ Actually Mean?

The matrix equation

$$N^n = 0$$

has a useful interpretation in terms of vectors.

For every vector v ,

$$N^n v = 0.$$

Thus repeated application of N must eventually send every vector to the zero vector.

For example, if

$$N^3 = 0,$$

then for every vector v ,

$$N^3 v = 0.$$

This does *not* mean that every vector requires exactly three applications of N before reaching zero.

Some vectors may already satisfy

$$Nv = 0.$$

Others may satisfy

$$Nv \neq 0, \quad N^2 v = 0,$$

while still others may satisfy

$$Nv \neq 0, \quad N^2v \neq 0, \quad N^3v = 0.$$

Therefore, the statement

$$N^3 = 0$$

should be interpreted as saying that three applications are *sufficient* for every vector, not that three applications are *necessary* for every vector.

Schematically, different vectors may behave as

$$v_1 \xrightarrow{N} 0,$$

or

$$v_2 \xrightarrow{N} v_1 \xrightarrow{N} 0,$$

or

$$v_3 \xrightarrow{N} v_2 \xrightarrow{N} v_1 \xrightarrow{N} 0.$$

This simple observation will later become one of the central ideas behind generalized eigenvectors and Jordan chains.

4.5 Why Cayley–Hamilton Matters for Our Development

The Cayley–Hamilton theorem does more than provide a convenient identity involving powers of a matrix. It tells us that the characteristic polynomial places an algebraic restriction on the action of the matrix itself.

In the repeated-eigenvalue example

$$p_A(\lambda) = (\lambda - 2)^3,$$

we know that

$$(A - 2I)^3 = 0.$$

At the same time, an ordinary eigenvector v_1 associated with $\lambda = 2$ satisfies

$$(A - 2I)v_1 = 0.$$

These two statements naturally raise an important question:

$(A - 2I)^3$ annihilates every vector,
but an eigenvector is annihilated after only one application.

What's the logic hiding here?

Investigating precisely these intermediate possibilities will lead us to generalized eigenvectors, nested null spaces, Jordan chains, and ultimately the Jordan canonical form.

5 From Ordinary Eigenvectors to Generalized Eigenvectors

We are now ready to begin the main development of the Jordan canonical form.

Suppose that λ is an eigenvalue of A , but its eigenspace does not contain enough linearly independent eigenvectors to construct a basis. Ordinary diagonalization then fails.

The question is not whether the missing directions disappear, but whether they can be recovered in a more general form.

Define

$$N = A - \lambda I.$$

An ordinary eigenvector $v \neq 0$ associated with λ satisfies

$$Nv = 0.$$

Hence the ordinary eigenspace is

$$E_\lambda = \ker(A - \lambda I) = \ker N.$$

Thus ordinary eigenvectors are precisely the nonzero vectors that are annihilated after one application of N .

5.1 Looking Beyond the Ordinary Eigenspace

To see where additional vectors may come from, consider first the special case in which the characteristic polynomial is

$$p_A(x) = (x - \lambda)^m.$$

By the Cayley–Hamilton theorem,

$$(A - \lambda I)^m = 0,$$

and therefore

$$\boxed{N^m = 0.}$$

Consequently,

$$N^m v = 0$$

for every vector v .

This is a much weaker condition than

$$Nv = 0.$$

An ordinary eigenvector is annihilated immediately, but other vectors may require several applications of N before reaching zero. This suggests studying the sequence

$$\boxed{\ker N \subseteq \ker N^2 \subseteq \ker N^3 \subseteq \cdots \subseteq \ker N^m.}$$

The reason these spaces are nested is simple. If

$$v \in \ker N^j,$$

then

$$N^j v = 0.$$

Applying N once more gives

$$N^{j+1} v = N(N^j v) = 0,$$

and hence

$$v \in \ker N^{j+1}.$$

Therefore, for every $j \geq 1$,

$$\ker N^j \subseteq \ker N^{j+1}.$$

These nested null spaces reveal structure that cannot be seen from the ordinary eigenspace alone.

Remark 5.1 (Do Not Reverse the Inclusion). *The implication*

$$N^j v = 0 \implies N^{j+1} v = 0$$

always holds, but the converse need not hold.

A vector may satisfy

$$N^{j+1} v = 0$$

while

$$N^j v \neq 0.$$

Thus

$$\ker N^j \subseteq \ker N^{j+1}$$

does not imply equality. The higher null spaces may contain genuinely new vectors.

The direction to remember is therefore

$$Nv = 0 \implies N^2 v = 0 \implies N^3 v = 0 \implies \dots,$$

and not the other way around.

5.2 Generalized Eigenvectors

The nested null spaces motivate an extension of the ordinary eigenvector concept.

Definition 5.2 (Generalized Eigenvector). *Let λ be an eigenvalue of A . A nonzero vector v is called a generalized eigenvector associated with λ if*

$$(A - \lambda I)^k v = 0$$

for some positive integer k .

Equivalently,

$$N^k v = 0$$

for some $k \geq 1$.

Ordinary eigenvectors are therefore included automatically: they are the special case $k = 1$.

More precisely, if k is the smallest positive integer for which

$$N^k v = 0,$$

then

$$N^{k-1} v \neq 0.$$

Such a vector is called a generalized eigenvector of *rank* k . Thus

$$\boxed{N^k v = 0, \quad N^{k-1} v \neq 0}$$

means that v requires exactly k applications of N to reach zero.

Equivalently,

$$\boxed{v \in \ker N^k \setminus \ker N^{k-1}.$$

This gives a natural hierarchy:

$$\begin{array}{ll} \ker N & : \text{ ordinary eigenvectors,} \\ \ker N^2 \setminus \ker N & : \text{ rank-2 generalized eigenvectors,} \\ \ker N^3 \setminus \ker N^2 & : \text{ rank-3 generalized eigenvectors,} \\ & \vdots \\ \ker N^k \setminus \ker N^{k-1} & : \text{ rank-}k \text{ generalized eigenvectors.} \end{array}$$

5.3 From Generalized Eigenvectors to a Chain

Generalized eigenvectors become especially useful when they are linked by successive applications of N .

Suppose v_k is a generalized eigenvector of rank k . Define

$$v_{k-1} = N v_k,$$

then

$$v_{k-2} = Nv_{k-1} = N^2v_k,$$

and continue in this way until

$$v_1 = N^{k-1}v_k.$$

Because v_k has rank k ,

$$v_1 \neq 0,$$

while

$$Nv_1 = N^k v_k = 0.$$

Hence v_1 is an ordinary eigenvector.

The resulting vectors satisfy

$$\boxed{Nv_1 = 0, \quad Nv_j = v_{j-1}, \quad j = 2, \dots, k.}$$

Schematically,

$$\boxed{v_k \xrightarrow{N} v_{k-1} \xrightarrow{N} \dots \xrightarrow{N} v_2 \xrightarrow{N} v_1 \xrightarrow{N} 0.}$$

Thus a rank- k generalized eigenvector naturally generates a sequence of vectors descending through the nested null spaces.

5.4 Jordan Chains

The preceding sequence is the central structure behind the Jordan canonical form.

Definition 5.3 (Jordan Chain). *Let λ be an eigenvalue of A . A sequence of nonzero vectors*

$$v_1, v_2, \dots, v_k$$

is called a Jordan chain of length k associated with λ if

$$\boxed{(A - \lambda I)v_1 = 0, \quad (A - \lambda I)v_j = v_{j-1}, \quad j = 2, \dots, k.}$$

Equivalently, with $N = A - \lambda I$,

$$Nv_1 = 0, \quad Nv_j = v_{j-1}, \quad j = 2, \dots, k.$$

The chain therefore has the form

$$\boxed{v_k \xrightarrow{N} v_{k-1} \xrightarrow{N} \cdots \xrightarrow{N} v_2 \xrightarrow{N} v_1 \xrightarrow{N} 0.}$$

The vector v_1 is an ordinary eigenvector, while v_j is a generalized eigenvector of rank j . Indeed,

$$N^j v_j = 0, \quad N^{j-1} v_j = v_1 \neq 0.$$

Therefore,

$$\boxed{v_j \in \ker N^j \setminus \ker N^{j-1}.}$$

A Jordan chain can thus be viewed as a sequence that connects the successive levels of the nested null spaces.

5.5 Why a Jordan Chain Produces a Jordan Block

The defining relations of a Jordan chain immediately determine how A acts on its vectors.

Since

$$N = A - \lambda I,$$

The first chain relation,

$$N v_1 = 0,$$

gives, after substituting

$$N = A - \lambda I,$$

the equation

$$(A - \lambda I) v_1 = 0.$$

Expanding,

$$A v_1 - \lambda I v_1 = 0.$$

Since

$$Iv_1 = v_1,$$

we obtain

$$Av_1 - \lambda v_1 = 0,$$

and therefore

$$\boxed{Av_1 = \lambda v_1.}$$

while

$$Nv_j = v_{j-1}, \quad j = 2, \dots, k,$$

gives, after substituting

$$N = A - \lambda I,$$

the relation

$$(A - \lambda I)v_j = v_{j-1}.$$

Expanding the left-hand side,

$$Av_j - \lambda Iv_j = v_{j-1}.$$

Since

$$Iv_j = v_j,$$

we obtain

$$Av_j - \lambda v_j = v_{j-1}.$$

Therefore,

$$\boxed{Av_j = \lambda v_j + v_{j-1}, \quad j = 2, \dots, k.}$$

Thus the action of A on the entire Jordan chain is

$$\begin{aligned}
Av_1 &= \lambda v_1, \\
Av_2 &= \lambda v_2 + v_1, \\
Av_3 &= \lambda v_3 + v_2, \\
&\vdots \\
Av_j &= \lambda v_j + v_{j-1}, \\
&\vdots \\
Av_k &= \lambda v_k + v_{k-1}.
\end{aligned}$$

Relative to the ordered chain

$$\mathcal{B} = (v_1, v_2, \dots, v_k),$$

these relations are represented by the matrix

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}.$$

This matrix is called the *Jordan block of size k* associated with the eigenvalue λ .

If the chain vectors form a basis of the space under consideration, let

$$P = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_k \\ | & | & \cdots & | \end{pmatrix}.$$

Then the chain relations give

$$AP = PJ_k(\lambda),$$

and therefore, provided P is invertible,

$$P^{-1}AP = J_k(\lambda).$$

The Jordan block is therefore not an arbitrary substitute for a diagonal matrix. It is the matrix representation naturally produced when A is expressed in a basis organized as a Jordan chain.

5.6 Where Do the Ones Come From?

The superdiagonal 1's in a Jordan block are a direct consequence of the chain relation

$$Av_j = \lambda v_j + v_{j-1}.$$

Relative to the ordered basis

$$\mathcal{B} = (v_1, v_2, \dots, v_k),$$

the vector Av_j therefore has coefficient λ in the v_j -direction and coefficient 1 in the v_{j-1} -direction.

Consequently,

$$\boxed{\lambda \text{ on the diagonal} \quad \longleftrightarrow \quad \text{the eigenvalue contribution,}}$$

whereas

$$\boxed{1 \text{ on the superdiagonal} \quad \longleftrightarrow \quad \text{the connection } v_j \mapsto v_{j-1}.}$$

Thus the 1's are not inserted merely by convention. They record the defining structure of the Jordan chain:

$$\boxed{(A - \lambda I)v_j = v_{j-1}.$$

This is the fundamental reason that a Jordan block has its characteristic form.

If the reader is still having difficulty seeing where the 1's come from, recall the basic rule of matrix–vector multiplication. The 1 selects the preceding vector v_{j-1} , while λ selects the current vector v_j , giving precisely

$$Av_j = v_{j-1} + \lambda v_j.$$

If this is still not evident, we recommend reviewing our other notes, where matrix–vector and matrix–matrix multiplication are discussed explicitly from several different perspectives.

6 Practical Intuition Behind Jordan Chains

So far, we have developed the mathematics behind generalized eigenvectors, Jordan chains, and Jordan blocks in considerable detail. We have used the Cayley–Hamilton theorem, nested null spaces, and the operator $A - \lambda I$ to explain how the Jordan structure arises.

We now temporarily put most of that formal mathematics aside.

The purpose of this section is to develop an intuitive picture of the same phenomenon through familiar situations. If the reader found the previous theoretical development difficult, it may actually be useful to begin here, develop the intuition, and then return to the preceding section. The algebra may look considerably more natural the second time around.

The central puzzle that might remain in our mind comes from comparing two facts.

In the single-eigenvalue situation considered earlier, the Cayley–Hamilton theorem tells us that, for some positive integer k ,

$$(A - \lambda I)^k v = 0$$

for every vector v .

Yet an ordinary eigenvector v_1 satisfies the much stronger condition

$$(A - \lambda I)v_1 = 0.$$

Something interesting is therefore happening. The operator may require several applications before an arbitrary vector is guaranteed to disappear, yet this particular vector disappears after only one.

Let us forget matrices for a moment and imagine an everyday situation.

6.1 The Medicine Analogy

Suppose a physician prescribes a medicine and determines that a patient may require several doses before the treatment is complete.

For the sake of the analogy, suppose that as many as k doses may be required.

We therefore expect something like

$$\text{patient} \longrightarrow \text{dose} \longrightarrow \text{dose} \longrightarrow \cdots \longrightarrow \text{recovery}.$$

But something surprising happens.

One particular patient becomes well immediately after the first dose.

Naturally, we would ask:

Why did this patient require only one dose when several doses might have been necessary?

One possible explanation is that the first dose we observed was not really the beginning of the story. Perhaps the patient had already received the same medicine through some other route. It might have been administered previously without our knowledge, or an equivalent substance might already have entered the patient's system through some other source.

The important point is not the medical details of the analogy. The important point is that an unexpectedly early outcome makes us suspect that there may be a *history behind what we are currently observing*.

This is precisely the kind of intuition that is useful for the Jordan construction.

The Cayley–Hamilton theorem tells us that repeated application of

$$A - \lambda I$$

will eventually annihilate the relevant vectors. But the ordinary eigenvector v_1 is already annihilated after one application:

$$(A - \lambda I)v_1 = 0.$$

Instead of merely accepting this as the end of the story, we may ask whether v_1 itself could have come from somewhere.

Could there be another vector v_2 such that

$$(A - \lambda I)v_2 = v_1?$$

If so, v_1 has what we may informally think of as a *parent*. And then another question immediately arises:

Could v_2 itself have a parent?

Perhaps there exists v_3 such that

$$(A - \lambda I)v_3 = v_2.$$

Continuing this way, what initially appeared to be a single ordinary eigenvector may turn out to be the visible end of a much longer history:

$$\boxed{v_k \longrightarrow v_{k-1} \longrightarrow \cdots \longrightarrow v_2 \longrightarrow v_1 \longrightarrow 0.}$$

The ordinary eigenvector v_1 is therefore not necessarily the beginning of the story. It may be the *last surviving member* of a sequence of vectors connected through repeated applications of $A - \lambda I$.

There is another useful way to visualize the same idea. Instead of thinking about medicine, consider the inheritance of a fatal genetic disease through a family.

For the purpose of this analogy, imagine that being sent to the null space—that is, being mapped to zero—corresponds to dying from the disease. A particular genetic condition may be carried through several generations without causing that outcome. A grandparent may carry the relevant genetic

variant, pass it to a parent, and the parent may in turn pass it to a child. Then, for biological reasons that may involve many additional factors, the disease may become severe in the child, and unfortunately they die.

The unexpected event naturally encourages the physicians to look backward through the family history. They may discover that what appeared to happen suddenly in the child actually has an ancestry:

$$\boxed{\text{grandparent} \longrightarrow \text{parent} \longrightarrow \text{child} \longrightarrow \text{death}.}$$

The child is where the outcome became visible, but the story did not necessarily begin with the child. Tracing backward reveals the generations that carried the relevant inherited factor.

We may use the same picture for our vectors. If reaching zero is thought of as reaching the “death” of the process, then

$$v_1 \xrightarrow{A-\lambda I} 0$$

shows us only the final visible step. We may therefore investigate whether v_1 has a parent v_2 , whether v_2 has a parent v_3 , and so on. Tracing this mathematical family tree backward may reveal

$$\boxed{v_k \xrightarrow{A-\lambda I} v_{k-1} \xrightarrow{A-\lambda I} \cdots \xrightarrow{A-\lambda I} v_2 \xrightarrow{A-\lambda I} v_1 \xrightarrow{A-\lambda I} 0.}$$

Of course, this is only an analogy: real genetic inheritance and disease expression are considerably more complicated. Its purpose here is simply to emphasize one idea. When we observe v_1 being sent immediately to zero, we can look backward and ask whether there is an entire ancestry of vectors behind it.

This sequence is exactly what the mathematics later identifies as a Jordan chain.

7 A Complete 3×3 Example

The theoretical development of the Jordan canonical form can appear somewhat convoluted when encountered for the first time. Fortunately, the computational problems themselves are usually much less difficult than the theory might suggest.

In a classroom or examination setting, a sensible problem designed to test the basic Jordan-form construction will ordinarily involve a small matrix. A 3×3 example is already large enough to reveal the essential structure: repeated eigenvalues, a shortage of ordinary eigenvectors, generalized eigenvectors, a Jordan chain, the Jordan block, and the similarity transformation.

There is therefore little pedagogical value in beginning with an unnecessarily large matrix. Instead, we will solve one carefully chosen 3×3 problem completely and step by step. This example will reveal the rest of the story developed in the previous sections.

Consider

$$A = \begin{pmatrix} 9 & 7 & 3 \\ -9 & -7 & -4 \\ 4 & 4 & 4 \end{pmatrix}.$$

Our goal is to find an invertible matrix P and a Jordan matrix J such that

$$P^{-1}AP = J.$$

We proceed one step at a time.

7.1 Step 1: Find the Characteristic Polynomial

We begin with

$$p_A(x) = \det(xI - A).$$

Since

$$xI = \begin{pmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & x \end{pmatrix},$$

we obtain

$$xI - A = \begin{pmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & x \end{pmatrix} - \begin{pmatrix} 9 & 7 & 3 \\ -9 & -7 & -4 \\ 4 & 4 & 4 \end{pmatrix}.$$

Therefore,

$$xI - A = \begin{pmatrix} x-9 & -7 & -3 \\ 9 & x+7 & 4 \\ -4 & -4 & x-4 \end{pmatrix}.$$

Hence

$$p_A(x) = \det \begin{pmatrix} x-9 & -7 & -3 \\ 9 & x+7 & 4 \\ -4 & -4 & x-4 \end{pmatrix}.$$

Expanding along the first row,

$$p_A(x) = (x-9) \begin{vmatrix} x+7 & 4 \\ -4 & x-4 \end{vmatrix} - (-7) \begin{vmatrix} 9 & 4 \\ -4 & x-4 \end{vmatrix} \\ + (-3) \begin{vmatrix} 9 & x+7 \\ -4 & -4 \end{vmatrix}.$$

Now compute each 2×2 determinant separately.

First,

$$\begin{aligned} \begin{vmatrix} x+7 & 4 \\ -4 & x-4 \end{vmatrix} &= (x+7)(x-4) - 4(-4) \\ &= (x+7)(x-4) + 16 \\ &= x^2 + 3x - 28 + 16 \\ &= x^2 + 3x - 12. \end{aligned}$$

Second,

$$\begin{aligned} \begin{vmatrix} 9 & 4 \\ -4 & x-4 \end{vmatrix} &= 9(x-4) - 4(-4) \\ &= 9x - 36 + 16 \\ &= 9x - 20. \end{aligned}$$

Third,

$$\begin{aligned} \begin{vmatrix} 9 & x+7 \\ -4 & -4 \end{vmatrix} &= 9(-4) - (x+7)(-4) \\ &= -36 + 4(x+7) \\ &= -36 + 4x + 28 \\ &= 4x - 8. \end{aligned}$$

Substituting these results,

$$p_A(x) = (x-9)(x^2 + 3x - 12) + 7(9x - 20) - 3(4x - 8).$$

Expand the first product:

$$\begin{aligned} (x-9)(x^2 + 3x - 12) &= x(x^2 + 3x - 12) - 9(x^2 + 3x - 12) \\ &= x^3 + 3x^2 - 12x - 9x^2 - 27x + 108 \\ &= x^3 - 6x^2 - 39x + 108. \end{aligned}$$

Also,

$$7(9x - 20) = 63x - 140,$$

and

$$-3(4x - 8) = -12x + 24.$$

Therefore,

$$\begin{aligned} p_A(x) &= x^3 - 6x^2 - 39x + 108 + 63x - 140 - 12x + 24 \\ &= x^3 - 6x^2 + 12x - 8. \end{aligned}$$

But

$$x^3 - 6x^2 + 12x - 8 = (x - 2)^3.$$

Hence

$$\boxed{p_A(x) = (x - 2)^3.}$$

Therefore the matrix has only one eigenvalue,

$$\boxed{\lambda = 2,}$$

with algebraic multiplicity

$$\boxed{3.}$$

Throughout this calculation, we have used x rather than λ as the variable in the characteristic polynomial simply for easier typesetting and readability. There is no mathematical difference between the two choices.

In this context, x and λ are only dummy variables used to represent the argument of the polynomial. Thus

$$p_A(x) = (x - 2)^3$$

and

$$p_A(\lambda) = (\lambda - 2)^3$$

express exactly the same characteristic polynomial. The reader should therefore not attach any special significance to the choice of symbol.

7.2 Step 2: Find the Ordinary Eigenvectors

We now solve

$$(A - 2I)v = 0.$$

First compute

$$A - 2I = \begin{pmatrix} 9 & 7 & 3 \\ -9 & -7 & -4 \\ 4 & 4 & 4 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Thus

$$N = A - 2I = \begin{pmatrix} 7 & 7 & 3 \\ -9 & -9 & -4 \\ 4 & 4 & 2 \end{pmatrix}.$$

Let

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Then

$$Nv = 0$$

becomes

$$\begin{pmatrix} 7 & 7 & 3 \\ -9 & -9 & -4 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} 7x + 7y + 3z &= 0, \\ -9x - 9y - 4z &= 0, \\ 4x + 4y + 2z &= 0. \end{aligned}$$

Since x and y always appear as $x + y$, let

$$s = x + y.$$

The system becomes

$$\begin{aligned}7s + 3z &= 0, \\ -9s - 4z &= 0, \\ 4s + 2z &= 0.\end{aligned}$$

From the third equation,

$$4s + 2z = 0.$$

Hence

$$2s + z = 0,$$

so

$$z = -2s.$$

Substitute this into the first equation:

$$7s + 3(-2s) = 0.$$

Thus

$$7s - 6s = 0,$$

and therefore

$$s = 0.$$

Hence

$$z = 0.$$

Since

$$s = x + y = 0,$$

we have

$$y = -x.$$

Therefore every eigenvector has the form

$$v = x \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

We choose

$$v_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

Thus

$$E_2 = \ker(A - 2I) = \text{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}.$$

Therefore,

$$\dim E_2 = 1.$$

The algebraic multiplicity is 3, but the geometric multiplicity is only 1:

$$1 < 3.$$

Hence A is defective and cannot be diagonalized.

This is exactly the situation for which the Jordan construction is needed.

7.3 Step 3: Search for the Parent of v_1

We have

$$Nv_1 = 0.$$

Following the intuition developed earlier, we now ask whether v_1 has a “parent.”

That is, can we find a vector v_2 satisfying

$$Nv_2 = v_1?$$

Let

$$v_2 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

Then

$$\begin{pmatrix} 7 & 7 & 3 \\ -9 & -9 & -4 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} 7a + 7b + 3c &= 1, \\ -9a - 9b - 4c &= -1, \\ 4a + 4b + 2c &= 0. \end{aligned}$$

Again let

$$s = a + b.$$

Then

$$\begin{aligned} 7s + 3c &= 1, \\ -9s - 4c &= -1, \\ 4s + 2c &= 0. \end{aligned}$$

From the third equation,

$$4s + 2c = 0.$$

Therefore,

$$c = -2s.$$

Substitute into the first equation:

$$7s + 3(-2s) = 1.$$

Hence

$$7s - 6s = 1,$$

so

$$s = 1.$$

Therefore,

$$c = -2.$$

Since

$$a + b = 1,$$

we are free to choose any convenient pair satisfying this relation.

Choose

$$a = 1, \quad b = 0.$$

Then

$$v_2 = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}.$$

Let us immediately verify this choice:

$$Nv_2 = \begin{pmatrix} 7 & 7 & 3 \\ -9 & -9 & -4 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}.$$

Carrying out the multiplication,

$$Nv_2 = \begin{pmatrix} 7(1) + 7(0) + 3(-2) \\ -9(1) - 9(0) - 4(-2) \\ 4(1) + 4(0) + 2(-2) \end{pmatrix}.$$

Thus

$$Nv_2 = \begin{pmatrix} 7 - 6 \\ -9 + 8 \\ 4 - 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

Therefore,

$$\boxed{Nv_2 = v_1.}$$

7.4 Step 4: Search for the Parent of v_2

We still need a third independent vector.

Following exactly the same idea, we now ask whether v_2 itself has a parent.

We seek v_3 satisfying

$$\boxed{Nv_3 = v_2.}$$

Let

$$v_3 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

Then

$$\begin{pmatrix} 7 & 7 & 3 \\ -9 & -9 & -4 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} 7a + 7b + 3c &= 1, \\ -9a - 9b - 4c &= 0, \\ 4a + 4b + 2c &= -2. \end{aligned}$$

Again define

$$s = a + b.$$

Then

$$\begin{aligned} 7s + 3c &= 1, \\ -9s - 4c &= 0, \\ 4s + 2c &= -2. \end{aligned}$$

From the second equation,

$$-9s - 4c = 0.$$

Therefore,

$$4c = -9s,$$

so

$$c = -\frac{9}{4}s.$$

Now substitute this into the third equation:

$$4s + 2\left(-\frac{9}{4}s\right) = -2.$$

Thus

$$4s - \frac{9}{2}s = -2.$$

Writing $4s$ as $\frac{8}{2}s$,

$$\frac{8}{2}s - \frac{9}{2}s = -2.$$

Therefore,

$$-\frac{1}{2}s = -2.$$

Multiplying both sides by -2 ,

$$s = 4.$$

Hence

$$c = -\frac{9}{4}(4),$$

so

$$c = -9.$$

Since

$$a + b = 4,$$

choose

$$a = 4, \quad b = 0.$$

Therefore,

$$\boxed{v_3 = \begin{pmatrix} 4 \\ 0 \\ -9 \end{pmatrix} .}$$

Again we verify directly:

$$Nv_3 = \begin{pmatrix} 7 & 7 & 3 \\ -9 & -9 & -4 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \\ -9 \end{pmatrix} .$$

Therefore,

$$Nv_3 = \begin{pmatrix} 7(4) + 7(0) + 3(-9) \\ -9(4) - 9(0) - 4(-9) \\ 4(4) + 4(0) + 2(-9) \end{pmatrix} .$$

Hence

$$Nv_3 = \begin{pmatrix} 28 - 27 \\ -36 + 36 \\ 16 - 18 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} .$$

Thus

$$\boxed{Nv_3 = v_2 .}$$

7.5 Step 5: We Have Found the Jordan Chain

We have now obtained

$$v_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 4 \\ 0 \\ -9 \end{pmatrix} .$$

They satisfy

$$Nv_1 = 0,$$

$$Nv_2 = v_1,$$

and

$$Nv_3 = v_2.$$

Since

$$N = A - 2I,$$

the chain is

$$\boxed{v_3 \xrightarrow{A-2I} v_2 \xrightarrow{A-2I} v_1 \xrightarrow{A-2I} 0.}$$

This is a Jordan chain of length 3.

7.6 Step 6: Recover the Action of A

From

$$(A - 2I)v_1 = 0,$$

we have

$$Av_1 - 2Iv_1 = 0.$$

Since

$$Iv_1 = v_1,$$

we obtain

$$Av_1 - 2v_1 = 0,$$

and hence

$$\boxed{Av_1 = 2v_1.}$$

Next,

$$(A - 2I)v_2 = v_1.$$

Therefore,

$$Av_2 - 2Iv_2 = v_1.$$

Since

$$Iv_2 = v_2,$$

we obtain

$$Av_2 - 2v_2 = v_1.$$

Hence

$$\boxed{Av_2 = 2v_2 + v_1.}$$

Finally,

$$(A - 2I)v_3 = v_2.$$

Therefore,

$$Av_3 - 2Iv_3 = v_2.$$

Since

$$Iv_3 = v_3,$$

we obtain

$$Av_3 - 2v_3 = v_2.$$

Thus

$$Av_3 = 2v_3 + v_2.$$

Altogether,

$$\begin{aligned} Av_1 &= 2v_1, \\ Av_2 &= v_1 + 2v_2, \\ Av_3 &= v_2 + 2v_3. \end{aligned}$$

7.7 Step 7: Construct the Jordan Matrix

The preceding relations immediately tell us what the matrix of A must look like in the ordered basis

$$\mathcal{B} = (v_1, v_2, v_3).$$

From

$$Av_1 = 2v_1,$$

the first coordinate column is

$$\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}.$$

From

$$Av_2 = v_1 + 2v_2,$$

the second coordinate column is

$$\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

From

$$Av_3 = v_2 + 2v_3,$$

the third coordinate column is

$$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

Therefore,

$$J = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

7.8 Step 8: Construct the Matrix P

The columns of P are the vectors of the Jordan chain in the order

$$v_1, v_2, v_3.$$

Therefore,

$$P = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix}.$$

Substituting the vectors,

$$P = \begin{pmatrix} 1 & 1 & 4 \\ -1 & 0 & 0 \\ 0 & -2 & -9 \end{pmatrix}.$$

Before using P^{-1} , we verify that P is invertible.

Compute

$$\det P = \det \begin{pmatrix} 1 & 1 & 4 \\ -1 & 0 & 0 \\ 0 & -2 & -9 \end{pmatrix}.$$

Expanding along the second row is convenient:

$$\det P = (-1)(-1)^{2+1} \begin{vmatrix} 1 & 4 \\ -2 & -9 \end{vmatrix}.$$

Thus

$$\det P = \begin{vmatrix} 1 & 4 \\ -2 & -9 \end{vmatrix}.$$

Therefore,

$$\det P = 1(-9) - 4(-2).$$

Hence

$$\det P = -9 + 8 = -1.$$

Thus

$$\boxed{\det P = -1 \neq 0.}$$

Therefore P is invertible.

7.9 Step 9: Verify $AP = PJ$

Before computing $P^{-1}AP$, it is useful to verify the equivalent relation

$$AP = PJ.$$

First compute AP :

$$AP = \begin{pmatrix} 9 & 7 & 3 \\ -9 & -7 & -4 \\ 4 & 4 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 4 \\ -1 & 0 & 0 \\ 0 & -2 & -9 \end{pmatrix}.$$

Compute the entries row by row.

For the first row,

$$\begin{aligned} (1,1) : & \quad 9(1) + 7(-1) + 3(0) = 2, \\ (1,2) : & \quad 9(1) + 7(0) + 3(-2) = 3, \\ (1,3) : & \quad 9(4) + 7(0) + 3(-9) = 9. \end{aligned}$$

For the second row,

$$\begin{aligned} (2,1) : & \quad -9(1) - 7(-1) - 4(0) = -2, \\ (2,2) : & \quad -9(1) - 7(0) - 4(-2) = -1, \\ (2,3) : & \quad -9(4) - 7(0) - 4(-9) = 0. \end{aligned}$$

For the third row,

$$\begin{aligned}(3, 1) : & \quad 4(1) + 4(-1) + 4(0) = 0, \\(3, 2) : & \quad 4(1) + 4(0) + 4(-2) = -4, \\(3, 3) : & \quad 4(4) + 4(0) + 4(-9) = -20.\end{aligned}$$

Therefore,

$$AP = \begin{pmatrix} 2 & 3 & 9 \\ -2 & -1 & 0 \\ 0 & -4 & -20 \end{pmatrix}.$$

Now compute PJ :

$$PJ = \begin{pmatrix} 1 & 1 & 4 \\ -1 & 0 & 0 \\ 0 & -2 & -9 \end{pmatrix} \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Again compute every entry.

For the first row,

$$\begin{aligned}(1, 1) : & \quad 1(2) + 1(0) + 4(0) = 2, \\(1, 2) : & \quad 1(1) + 1(2) + 4(0) = 3, \\(1, 3) : & \quad 1(0) + 1(1) + 4(2) = 9.\end{aligned}$$

For the second row,

$$\begin{aligned}(2, 1) : & \quad (-1)(2) + 0(0) + 0(0) = -2, \\(2, 2) : & \quad (-1)(1) + 0(2) + 0(0) = -1, \\(2, 3) : & \quad (-1)(0) + 0(1) + 0(2) = 0.\end{aligned}$$

For the third row,

$$\begin{aligned}(3, 1) : & \quad 0(2) + (-2)(0) + (-9)(0) = 0, \\(3, 2) : & \quad 0(1) + (-2)(2) + (-9)(0) = -4, \\(3, 3) : & \quad 0(0) + (-2)(1) + (-9)(2) = -20.\end{aligned}$$

Therefore,

$$PJ = \begin{pmatrix} 2 & 3 & 9 \\ -2 & -1 & 0 \\ 0 & -4 & -20 \end{pmatrix}.$$

Hence

$$\boxed{AP = PJ.}$$

This already verifies that the Jordan-chain construction is correct.

7.10 Step 10: Verify $P^{-1}AP = J$

For completeness, we now verify the similarity transformation directly.

For this matrix,

$$P^{-1} = \begin{pmatrix} 0 & -1 & 0 \\ 9 & 9 & 4 \\ -2 & -2 & -1 \end{pmatrix}.$$

We already found

$$AP = \begin{pmatrix} 2 & 3 & 9 \\ -2 & -1 & 0 \\ 0 & -4 & -20 \end{pmatrix}.$$

Therefore,

$$P^{-1}AP = \begin{pmatrix} 0 & -1 & 0 \\ 9 & 9 & 4 \\ -2 & -2 & -1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 9 \\ -2 & -1 & 0 \\ 0 & -4 & -20 \end{pmatrix}.$$

Compute the first row:

$$\begin{aligned} (1, 1) : & \quad 0(2) + (-1)(-2) + 0(0) = 2, \\ (1, 2) : & \quad 0(3) + (-1)(-1) + 0(-4) = 1, \\ (1, 3) : & \quad 0(9) + (-1)(0) + 0(-20) = 0. \end{aligned}$$

Compute the second row:

$$\begin{aligned} (2, 1) : & \quad 9(2) + 9(-2) + 4(0) = 0, \\ (2, 2) : & \quad 9(3) + 9(-1) + 4(-4) = 2, \\ (2, 3) : & \quad 9(9) + 9(0) + 4(-20) = 1. \end{aligned}$$

Compute the third row:

$$\begin{aligned}
(3, 1) : \quad & (-2)(2) + (-2)(-2) + (-1)(0) = 0, \\
(3, 2) : \quad & (-2)(3) + (-2)(-1) + (-1)(-4) = 0, \\
(3, 3) : \quad & (-2)(9) + (-2)(0) + (-1)(-20) = 2.
\end{aligned}$$

Therefore,

$$P^{-1}AP = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Hence,

$$P^{-1}AP = J = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

7.11 What This Example Reveals

This single 3×3 problem contains nearly the entire basic story of the Jordan canonical form.

The characteristic polynomial told us that the eigenvalue 2 appears three times:

$$p_A(x) = (x - 2)^3.$$

However, the ordinary eigenspace supplied only one independent eigenvector:

$$v_1.$$

Ordinary diagonalization therefore failed.

Rather than abandoning the problem, we traced backward:

$$Nv_2 = v_1,$$

and then

$$Nv_3 = v_2.$$

Thus the missing directions were recovered through the Jordan chain

$$v_3 \longrightarrow v_2 \longrightarrow v_1 \longrightarrow 0.$$

In the language of our earlier intuition, v_1 was only the final visible member of a larger mathematical family. Searching for its parent and grandparent recovered the vectors that ordinary diagonalization could not provide.

Those vectors then formed the basis

$$P = [v_1 \ v_2 \ v_3],$$

and in that basis the complicated matrix

$$A = \begin{pmatrix} 9 & 7 & 3 \\ -9 & -7 & -4 \\ 4 & 4 & 4 \end{pmatrix}$$

became the remarkably simple matrix

$$J = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

The diagonal entries preserve the eigenvalue 2, while the superdiagonal 1's preserve the connections between successive members of the Jordan chain.

Thus Jordan canonical form does not somehow manufacture the missing eigenvectors. Instead, it replaces the unavailable ordinary eigendirections with generalized eigenvectors that retain the remaining algebraic structure of the transformation.

8 Computing the Jordan Canonical Form in MATLAB

After working through the Jordan construction by hand, it is useful to see how the same computation can be performed in MATLAB.

MATLAB provides the function

```
J = jordan(A)
```

for computing the Jordan canonical form. If we also want the transformation matrix, we use

```
[P,J] = jordan(A)
```

where the output satisfies

$$P^{-1}AP = J.$$

The columns of P form the corresponding Jordan basis, consisting of eigenvectors and, when necessary, generalized eigenvectors.

8.1 Our 3×3 Example in MATLAB

Consider again

$$A = \begin{pmatrix} 9 & 7 & 3 \\ -9 & -7 & -4 \\ 4 & 4 & 4 \end{pmatrix}.$$

The basic MATLAB calculation is remarkably short:

```
clear; clc;

A = [ 9  7  3;
     -9 -7 -4;
       4  4  4];

[P,J] = jordan(A);

disp('Transformation matrix P:')
disp(P)

disp('Jordan canonical form J:')
disp(J)
```

For this problem, the Jordan matrix has the structure

$$J = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

The matrix P returned by MATLAB need not contain exactly the same generalized eigenvectors that we constructed by hand. Generalized eigenvectors are not unique. What matters is that the returned matrices satisfy

$$P^{-1}AP = J.$$

A convenient verification is obtained from the equivalent relation

$$AP = PJ.$$

Thus we may check

$A*P - P*J$

and verify that the result is the zero matrix.

We may also directly examine

$\text{inv}(P)*A*P$

and compare the result with J .

For learning purposes, this is essentially all that is required: construct A , call `jordan`, examine P and J , and verify the decomposition.

8.2 Numerical Consideration

The preceding discussion concerns the exact algebraic structure of a matrix. In a textbook problem, the entries of A are assumed to be known exactly, and questions such as algebraic multiplicity, geometric multiplicity, and Jordan structure are therefore perfectly well defined.

Matrices arising in scientific computation, however, are often less idealized. They may come from experimental measurements, parameter estimation, numerical simulation, discretization of a differential equation or partial differential equation, or previous floating-point calculations. Consequently, the matrix actually stored in a computer may differ slightly from the ideal mathematical matrix.

A convenient way to represent this is

$$A_\varepsilon = A + \varepsilon E,$$

where ε is small and E describes the direction of the perturbation.

The term

$$\varepsilon E$$

may be interpreted as a small amount of unavoidable impurity introduced by measurement, modeling, discretization, or finite-precision computation.

Why This Matters for Jordan Form

Jordan structure is extremely sensitive to perturbations.

Suppose the exact matrix A has a repeated eigenvalue

$$2, 2, 2$$

and contains a nontrivial Jordan block.

A nearby matrix

$$A_\varepsilon = A + \varepsilon E$$

may instead have three slightly different eigenvalues:

$$\boxed{2, 2, 2 \longrightarrow 2 + \delta_1, 2 + \delta_2, 2 + \delta_3,}$$

where the quantities $\delta_1, \delta_2, \delta_3$ are small but generally unequal.

If the perturbed eigenvalues become distinct, the nearby matrix becomes diagonalizable.

Thus a matrix that is exactly defective in theory may appear nondefective when represented by real data or floating-point numbers.

This is not a contradiction. The two matrices are simply not algebraically identical.

Impurity in Real Data

Small perturbations may arise naturally from

- measurement uncertainty,
- sensor noise,
- parameter estimation,
- floating-point representation,
- rounding error,
- discretization error,
- or previous numerical computation.

Therefore, even if the ideal mathematical model contains an exact Jordan block, the matrix available in a computer may instead be a nearby matrix having a complete set of ordinary eigenvectors.

In some numerical calculations, this may make the nearby problem easier to manipulate.

The important distinction is

$$\boxed{\text{numerically easier} \neq \text{algebraically identical.}}$$

The nearby matrix may be simpler computationally while no longer possessing the exact Jordan structure of the original matrix.

8.3 A Simple MATLAB Experiment

This sensitivity can be demonstrated directly in MATLAB.

Consider again

```
clear; clc;

A = [ 9  7  3;
     -9 -7 -4;
       4  4  4];
```

The exact matrix has the repeated eigenvalue 2.

Now suppose a small perturbation is already present:

```
rng(1);

epsilon = 1e-8;
E = randn(size(A));

Apert = A + epsilon*E;

disp('Eigenvalues of the original matrix:')
disp(eig(A))

disp('Eigenvalues of the perturbed matrix:')
disp(eig(Apert))
```

The eigenvalues of the perturbed matrix will generally no longer be exactly identical.

Even though

$$\|A_\varepsilon - A\|$$

is small, the algebraic classification of the matrix may change.

This illustrates one of the central numerical difficulties associated with Jordan form:

small changes in the matrix can produce large changes in its Jordan structure.

8.4 Exact Algebra Versus Floating-Point Computation

The Jordan canonical form therefore provides an excellent example of the distinction between exact linear algebra and numerical linear algebra.

In exact mathematics, one asks

What is the Jordan structure of A ?

In numerical computation, one must also ask

How reliably can that structure be detected from the available data?

A nontrivial Jordan block represents an exact algebraic relationship. Because such relationships can be destroyed by very small perturbations, they are difficult to identify reliably from approximate floating-point data.

This is one reason that Jordan canonical form is extremely important theoretically but is used with caution in numerical computation.

8.5 A Changing Computational Landscape

There is, however, an important broader perspective.

For several decades, mainstream scientific computing was dominated by floating-point numerical methods, while exact symbolic manipulation played a more specialized role.

In recent years, symbolic computation has gained renewed momentum. Modern computer algebra systems, faster hardware, automatic algebraic manipulation, exact arithmetic, and hybrid symbolic–numerical methods have made exact computation practical in a much wider range of problems.

This development is particularly relevant to questions such as Jordan form, because the major difficulty often arises not from the algebra itself, but from representing exact algebraic structure using approximate numbers.

As symbolic and symbolic–numerical methods continue to improve, some of the computational difficulties currently associated with detecting exact Jordan structure may become less severe.

It is therefore reasonable to expect that future computational tools may handle certain defective–matrix problems more naturally than present floating-point methods do.

For now, however, the essential lesson is

exact algebra reveals the Jordan structure,
floating-point perturbations may obscure or destroy it,
and symbolic computation offers a possible bridge between the two.

The Jordan canonical form is therefore not only a classical theoretical construction. It is also a useful example of how the meaning of a mathematical object can depend strongly on the computational framework used to represent it.

9 Conclusion

We began this note with one of the most familiar ideas in linear algebra: diagonalization. When a matrix possesses enough linearly independent eigenvectors, the situation is remarkably clean. We form a basis from those eigenvectors and obtain

$$P^{-1}AP = D,$$

where D is diagonal.

The difficulty appears when the eigenvectors run out.

A repeated eigenvalue may have algebraic multiplicity larger than the dimension of its eigenspace, leaving us without enough ordinary eigenvectors to construct a basis. At first sight, this appears to be the end of the diagonalization procedure.

But mathematics rarely progresses by simply declaring defeat when the most obvious method stops working.

Instead, we changed the question.

Rather than asking only

Where does a vector go under $A - \lambda I$?

we also began asking

Where could this vector have come from?

That small change in perspective opened the door to the entire Jordan construction.

Starting from an ordinary eigenvector

$$(A - \lambda I)v_1 = 0,$$

we searched backward for a vector v_2 satisfying

$$(A - \lambda I)v_2 = v_1.$$

Then we asked whether v_2 itself had a predecessor:

$$(A - \lambda I)v_3 = v_2.$$

Continuing in this manner produced the chain

$$v_k \longrightarrow v_{k-1} \longrightarrow \cdots \longrightarrow v_2 \longrightarrow v_1 \longrightarrow 0.$$

What initially appeared to be a shortage of eigenvectors was therefore not the absence of structure. The structure was simply deeper than ordinary eigenvectors alone could reveal.

This is one of the broader lessons of the Jordan canonical form.

When the Forward Search Fails, Look Backward

Much of elementary mathematics trains us to think forward:

given something $\longrightarrow$ compute what comes next.

That is an enormously useful habit, but it is not the only way mathematicians think.

When a forward search becomes blocked, it can be fruitful to reverse the direction of the question.

Instead of asking only

What does this object produce?

one may ask

What could have produced this object?

The Jordan chain is a particularly elegant example of this principle.

The ordinary eigenvector v_1 appears to be the end of the story because

$$(A - \lambda I)v_1 = 0.$$

But by searching backward, we ask whether v_1 might itself be the image of some v_2 , whether v_2 might be the image of some v_3 , and so forth.

The apparent endpoint becomes the starting point of a new investigation.

This style of reasoning extends far beyond Jordan canonical form. Mathematical problem solving often involves changing direction, reversing a map, reconstructing causes from consequences, or asking what hidden structure must exist in order to produce what we already observe.

Broadening Intuition

There is another lesson worth carrying beyond this particular topic.

At some point in mathematical study, intuition must become broader than formal manipulation alone.

Proofs are indispensable. Computation is indispensable. Precise definitions are indispensable. But when a subject becomes buried under increasingly abstruse notation or lengthy algebra, repeatedly pushing symbols around may not be the best way to gain understanding.

Sometimes one should temporarily put the formalism aside and ask:

What is actually happening here?

That was the purpose of the practical analogies used in this note. They were not substitutes for proof. Rather, they provided another way of seeing the structure that the proof was describing.

Once the underlying picture becomes visible, one can return to the formal mathematics with a much better sense of why each equation is there.

A useful habit is therefore

$\text{formal theory} \longrightarrow \text{intuition} \longrightarrow \text{return to the formal theory.}$

The second encounter with the mathematics is often much clearer than the first.

The Jordan canonical form therefore leaves us with a lesson that is larger than the decomposition itself. When a familiar method reaches an obstruction, the answer is not always to push the same calculation harder. Sometimes progress comes from changing the question, reversing the direction of thought, or enlarging the framework in which the problem is being viewed.

In our case, the failure of ordinary diagonalization was not the end of the story. It was precisely the obstruction that led us to generalized eigenvectors, Jordan chains, and ultimately a richer understanding of the transformation.

References

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